

MATH 180 Strategies of Problem Solving

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Outline

- 1 SoPS Lecture 7
 - Getting started
 - Firing algorithms
 - Wrapping up

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Review

Last time

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- we played Battleship

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- we learned probability

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Last time

- we played Battleship
- we learned probability
- we used probability in battleship

Making groups

All homework will be done in groups

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- form groups of 3-5 people

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- sign up on Canvas!

Making groups

All homework will be done in groups

Groups

- form groups of 3-5 people
- sign up on Canvas!
- only one person per group submits the assignment

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Static firing algorithms

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Vocabulary

A **static firing algorithm** is one which fires in a pre-determined pattern, not changing whether each shot is a hit or a miss.

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A **static firing algorithm** is one which fires in a pre-determined pattern, not changing whether each shot is a hit or a miss.

Vocabulary

A **dynamic firing algorithm** is one where the algorithm can take into account the current board state before choosing each shot.

Static Examples

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Static example:

Fire the first shot at A1, the second shot at A2, the third shot at A3, and so on, proceeding left to right and top to bottom like reading a book.

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Static example:

Randomly choose a spot on the board and fire at it.

Dynamic Example

Dynamic example:

Fire the first shot at E5. For each subsequent shot, do the following:

- if you got a hit but didn't sink, continue a generic sinking algorithm
- otherwise, use a Monte Carlo method on a computer with 1000 samples to approximate the probabilities of each space being occupied. Fire at the most probable spot. If multiple spots have the same probability, randomly choose between them.

Best static algorithm

Best static algorithm

Problem:

Can we find the best static algorithm?

Best static algorithm

Problem:

Can we find the best static algorithm?

Minor adjustment:

We'll assume that after we get a hit, we run a generic sinking routine, and then return to the next available spot to fire at in our static algorithm. (Technically, this makes this not quite a static algorithm).

Battleship simulator

Activity

Use the [Battleship Simulator \(link\)](#) to explore different static firing patterns. Try to find the pattern that averages the least shots, averaged over thousands of simulations.

Probability heat map

Heat map

A **heat map** is a method of visualizing data that represents different values with colors. We can use a heat map to visualize the probability distribution of the battleship fleet at different points in the grid at various stages in the game.

Activity

Use the [Battleship Heatmap \(link\)](#) to explore different probabilities at different states of the game.

Battleship simulator

Activity

Using both tools in unison, try to design a best possible static algorithm.

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Static vs Dynamic

A **dynamic** firing algorithm is one where the shots change depending on what happens on the board.

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Question 1

What makes a good static firing algorithm?

Discussion

Static vs Dynamic

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Question 1

What makes a good static firing algorithm?

Question 2

Can you think of an interesting dynamic firing algorithm that might improve over the static algorithms we found?

Question of the Day

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Puzzle

What should the sum of the probabilities of each position on the board be, and why?

Final thoughts

Reflection

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- What was the static pattern you found that worked the best?

Final thoughts

Reflection

- What was the static pattern you found that worked the best?
- What dynamic algorithm would you suggest to improve your static algorithm.

Submit your responses by **Tonight!**